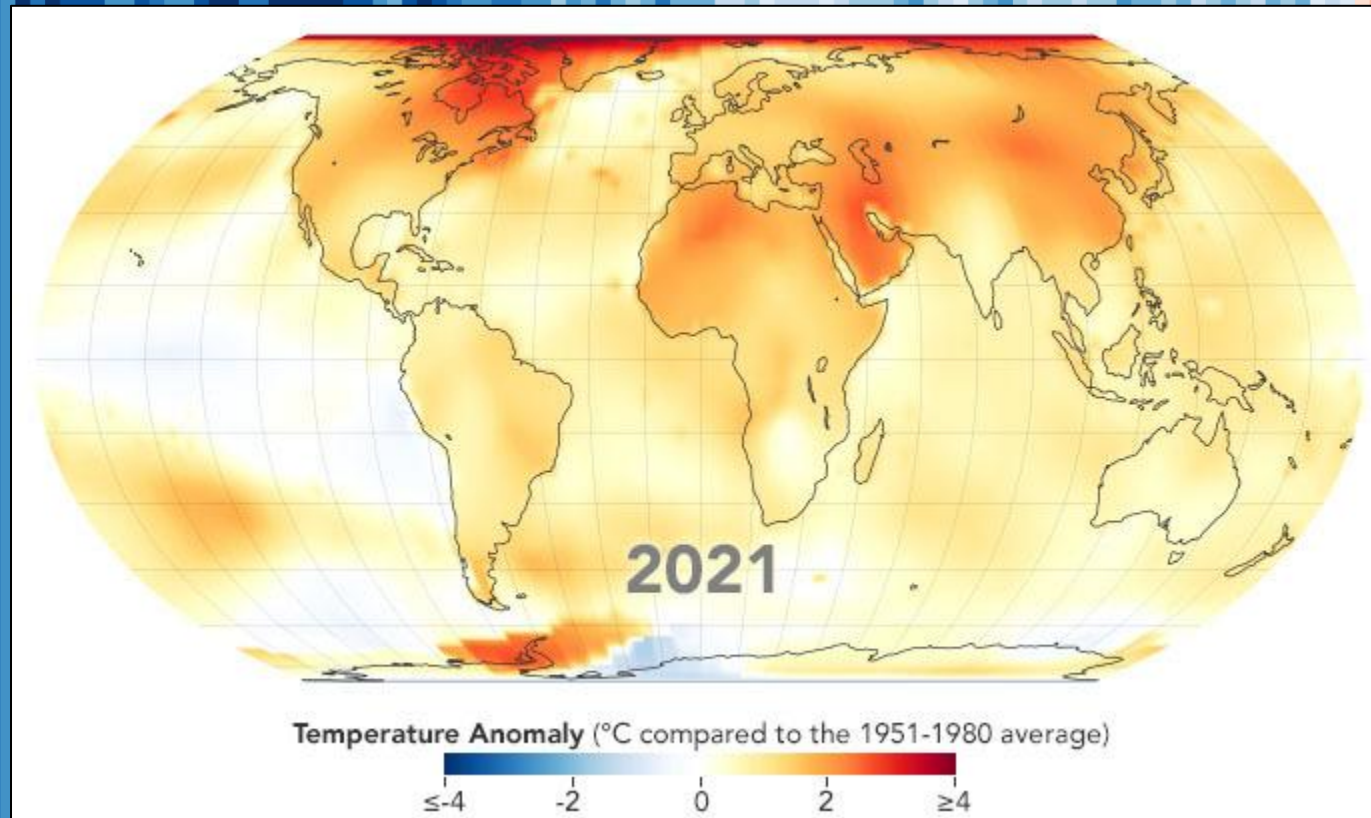


Climate & Us

Week 2: Past, Present, and Future Climate





Learning Objectives

Frontiers of Science

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Learning Objectives

Frontiers of Science

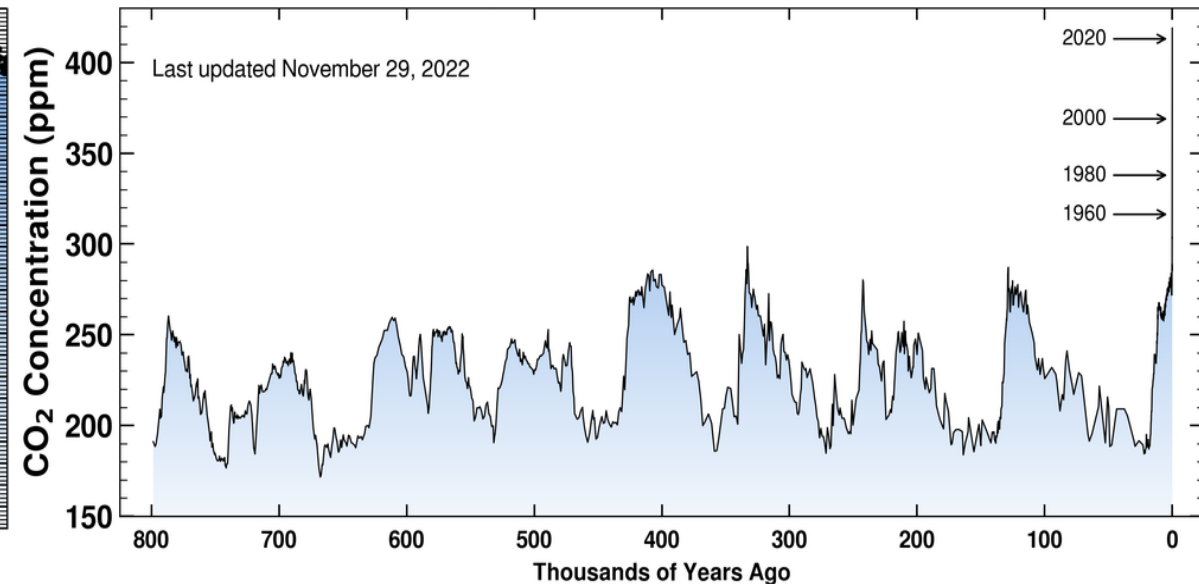
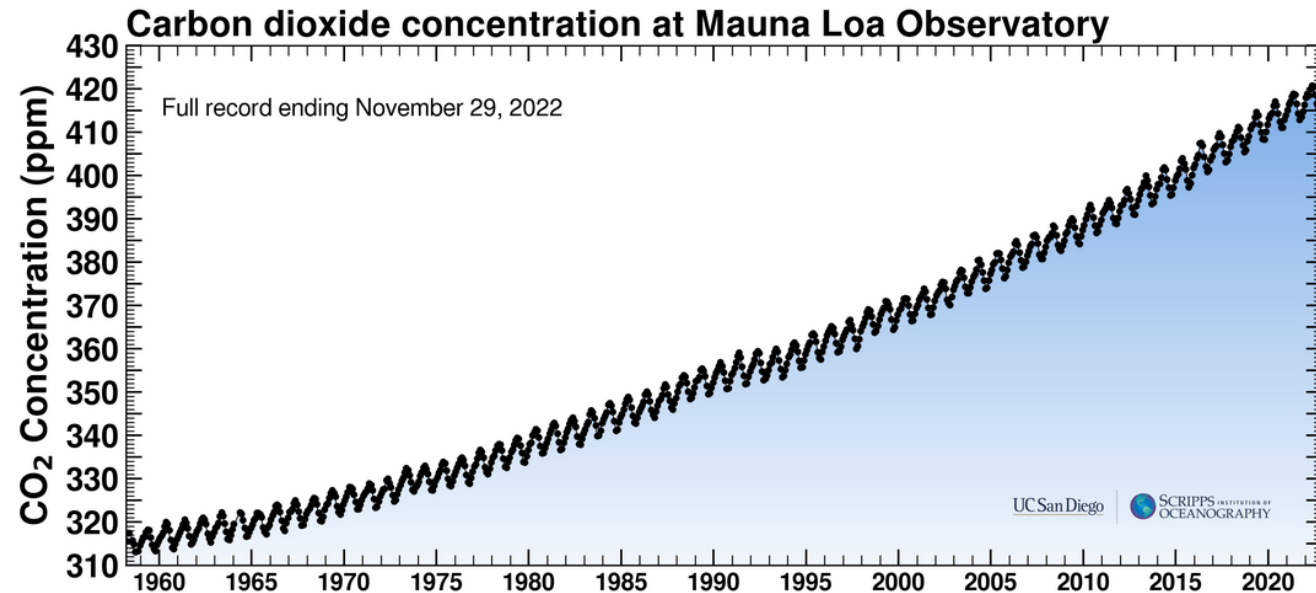
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LO1: Past climate

Frontiers of Science

- Data from modern observatories: CO₂ has increased dramatically over the last 60 years
- How do we place this change in historical context?

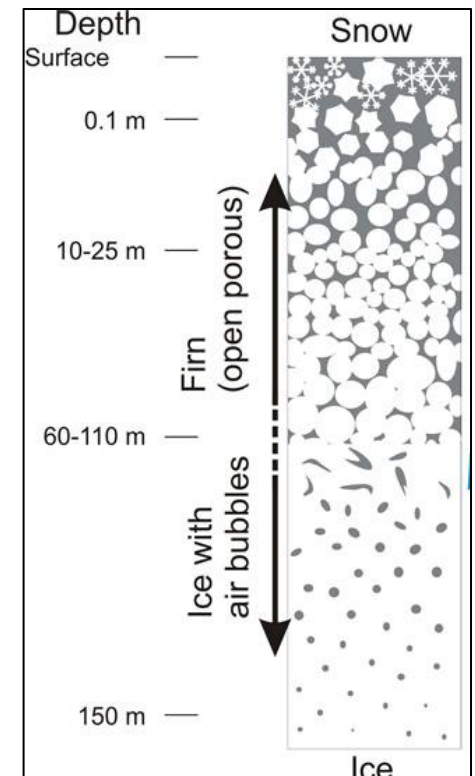
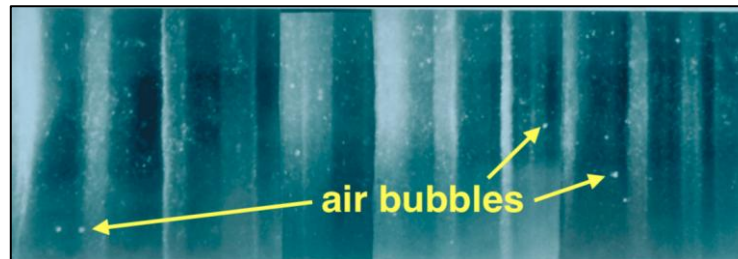
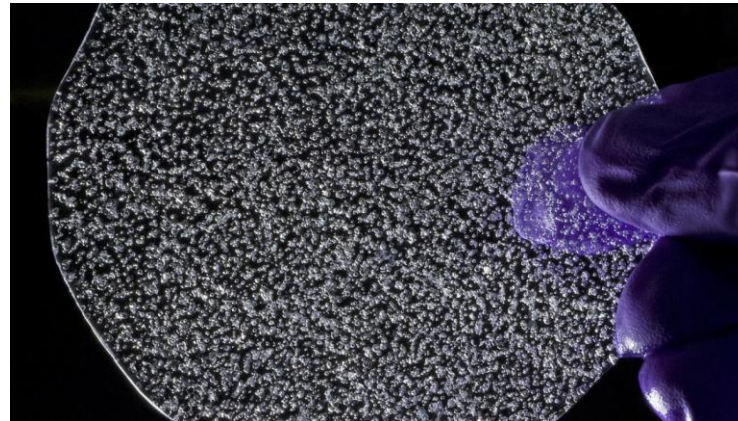




LO1: Paleoclimate records: Ice Cores

Frontiers of Science

- Ice cores: trapped air bubbles sample atmosphere each year when it snows
 - Air bubbles can be used to directly determine CO₂ concentration and C isotopes
 - Thickness of annual layers can be used to infer past temperature (qualitative proxy)
 - Oxygen and Hydrogen isotopes are proxies for temperature

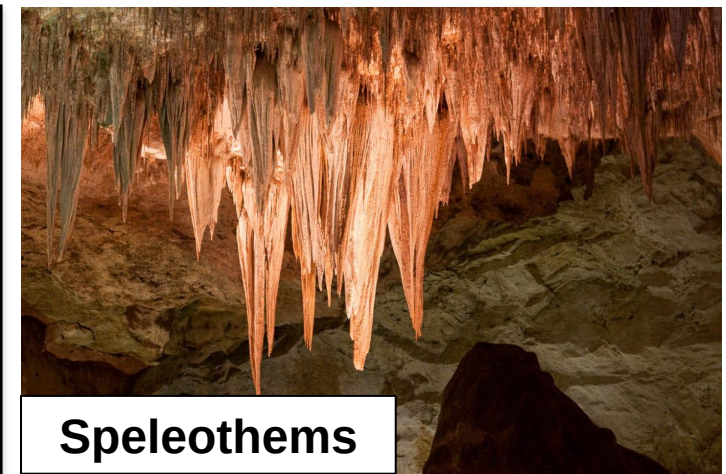
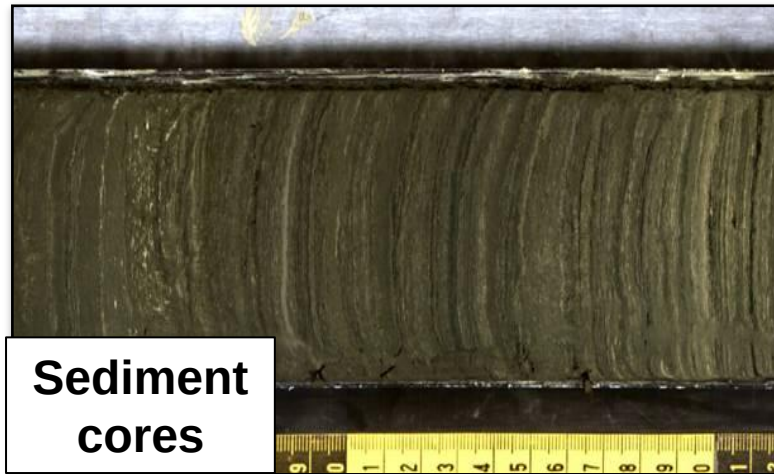




LO1: Paleoclimate records

Frontiers of Science

- Measurements of air composition in ice cores is one of several methods that researchers use to understand paleoclimate conditions





LO1: Paleoclimate records

- Different records provide estimates across different time scales

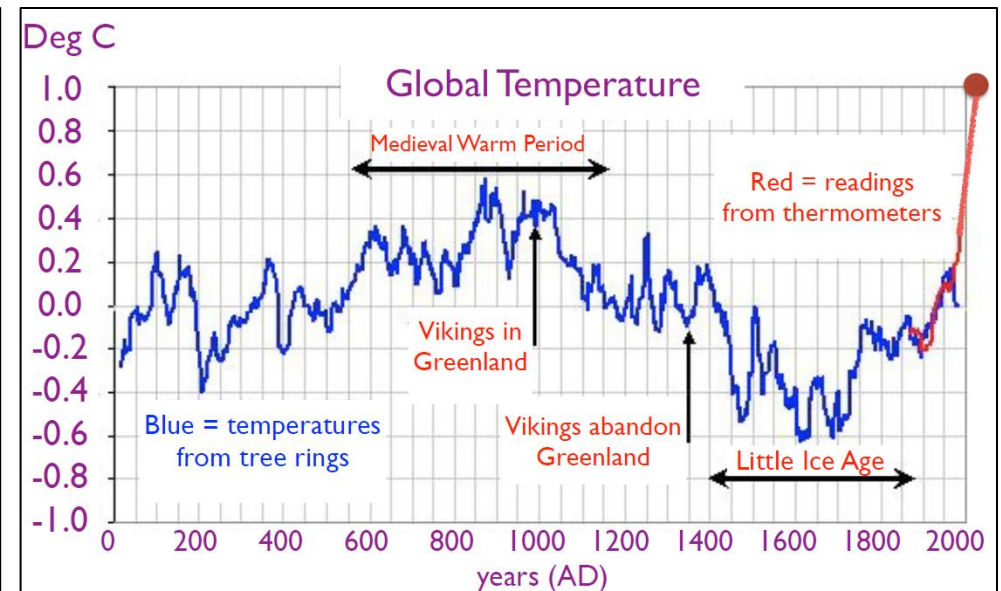
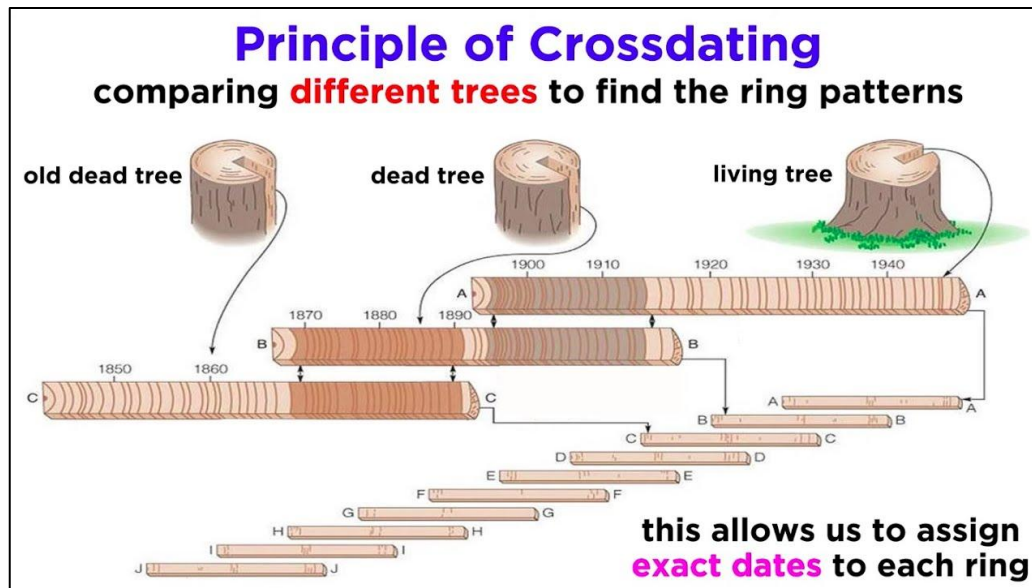
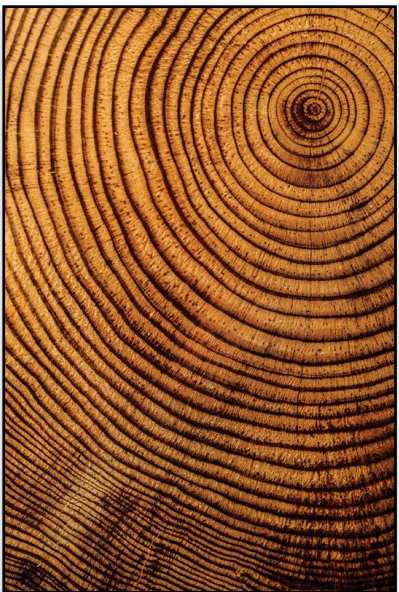
Record	Timescale
Instrumental measurements	150 years
Written historical records	~1500 years
Tree rings	13,910 years
Ice cores	800,000 years
Ocean floor sediments	50 million years
Speleothems	290 million years
Fossils and rocks	4.57 billion years



LO1: Paleoclimate records: Tree Rings

Frontiers of Science

- Tree rings record conditions each year
 - Oldest trees (Bristlecone pines): 5,000 yrs old, but we can correlate ring pattern & isotope records from different trees with overlap= record for 13,910 yrs
 - Thickness is a proxy for [relative] temperature / precipitation
 - Oxygen isotopes ($^{18}\text{O}/^{16}\text{O}$) is proxy for temperature
 - Carbon isotopes ($^{13}\text{C}/^{12}\text{C}$) is proxy for precipitation

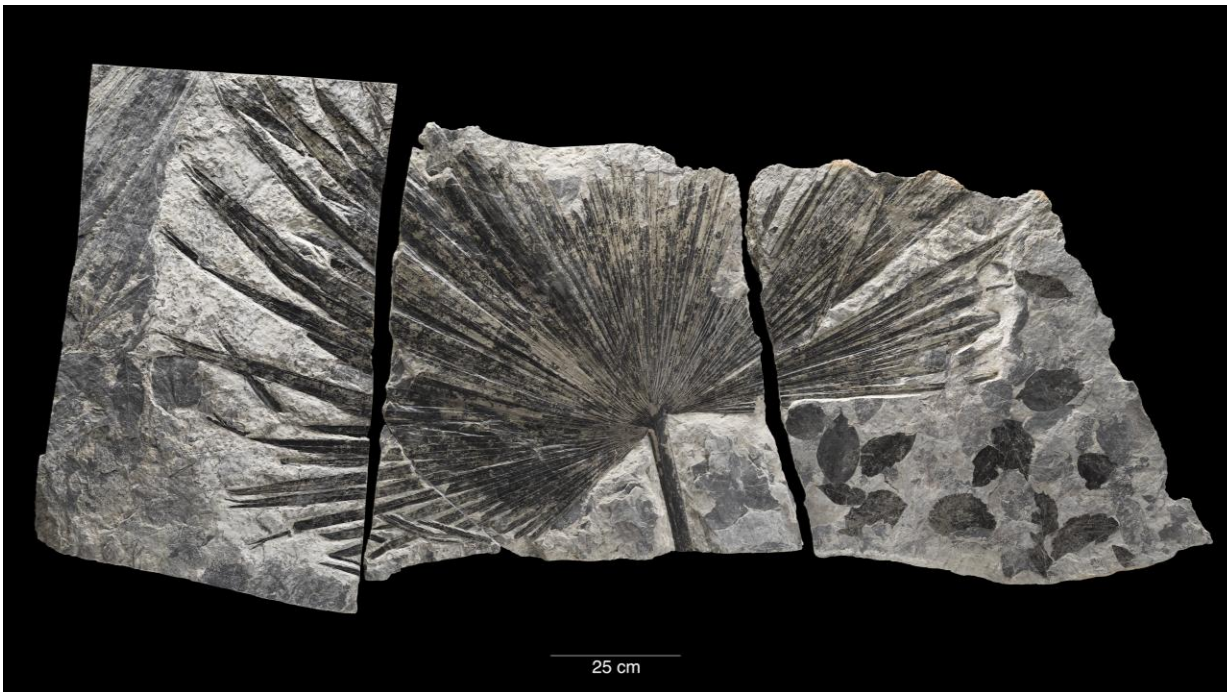




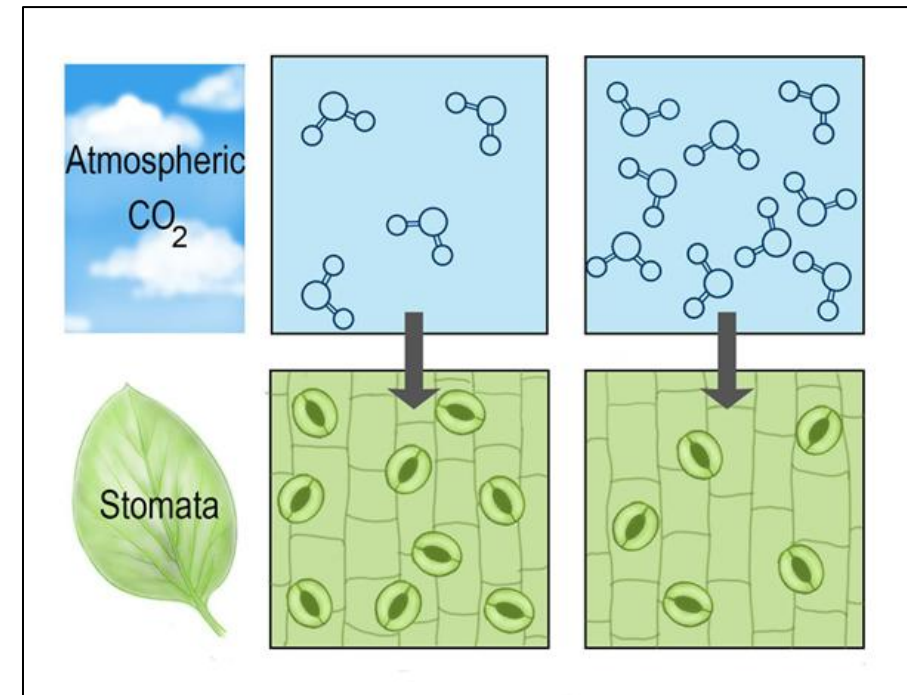
LO1: Paleoclimate records: Fossil Plants

Frontiers of Science

- Stomatal density on fossil leaves can be used to infer past CO₂ concentrations
 - Stomata: pores that help the plants take in CO₂
 - lower CO₂ concentrations → harder to get CO₂ → more stomata



Fossilized leaves from Antarctica- 60 Myrs ago (Smithsonian)



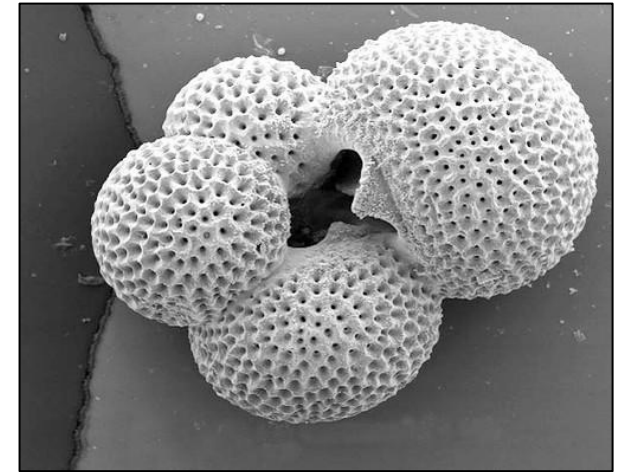
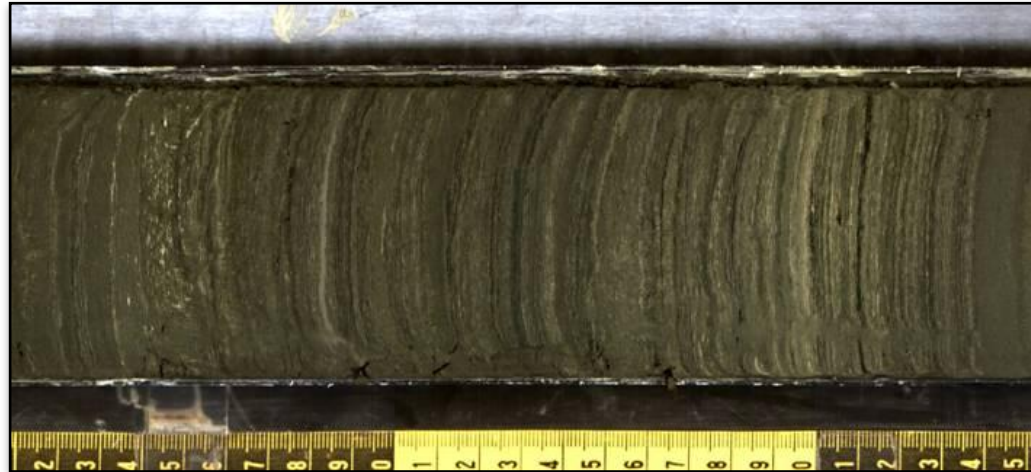
<https://www.nrm.se/english/researchandcollections/palaeobiology/ourresearch/globalginkgo.9005695.html>



LO1: Paleoclimate records: Sediment Cores

Frontiers of Science

- Organic matter deposits in layers on ocean or lake floor (oldest on bottom)
 - O and H Isotope ratios from phytoplankton shells- proxy for temperature
 - Pollen grains can be used to infer species distributions- proxy for climate





LO1: Paleoclimate records: Speliothems

Frontiers of Science

- Layers of CaCO_3 are deposited on stalactites/mites (“speliothems”) in caves
 - Water with dissolved O, C, and Ca (isotopes from atmosphere) seeps down into caves
 - Can use O, C, Ca isotopes in each growth band as proxy for T and precipitation



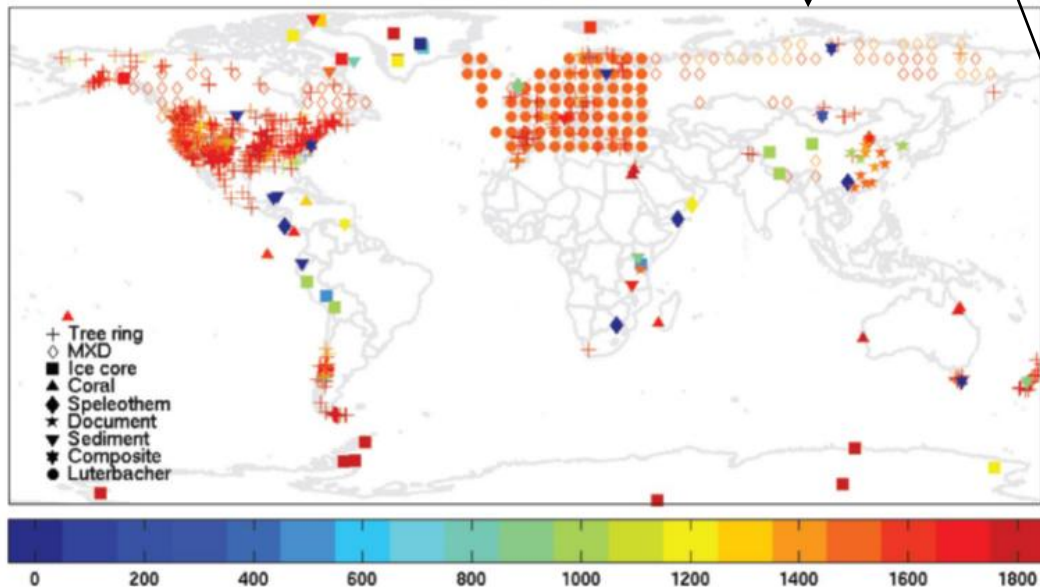


Paleoclimate Records- Converging Evidence/ Proxies

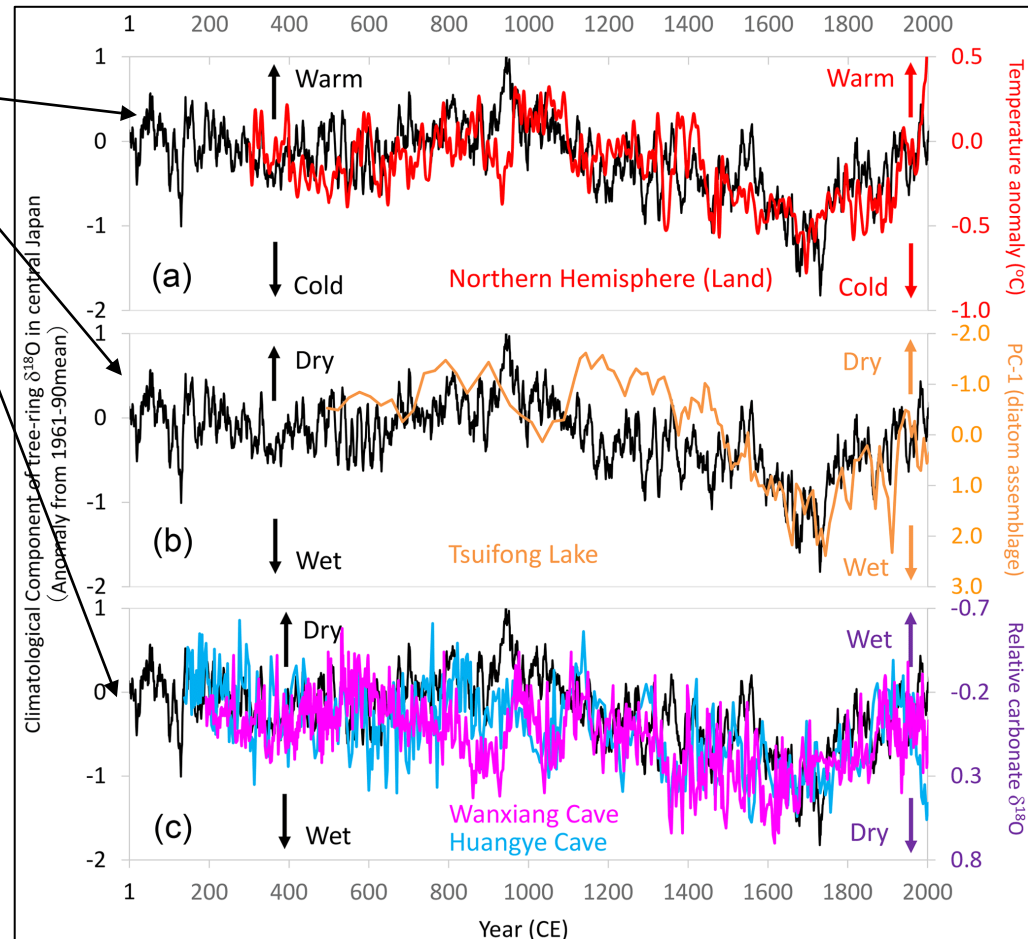
Frontiers of Science

- Scientists use multiple paleoclimate records and proxies to provide converging evidence of past climate!

Black: Temp
--determined from 9 different
proxies, samples worldwide



Mann et al., 2008



Japan:
Tree ring $^{18}\text{O}/^{16}\text{O}$
Proxy for T

SE China lake:
Plankton shells
Proxy for precip.

NW China caves
Speleothem $^{18}\text{O}/^{16}\text{O}$
Proxy for precip.

Nakatsuka et al., 2020



LO1: Comparing paleoclimate to today

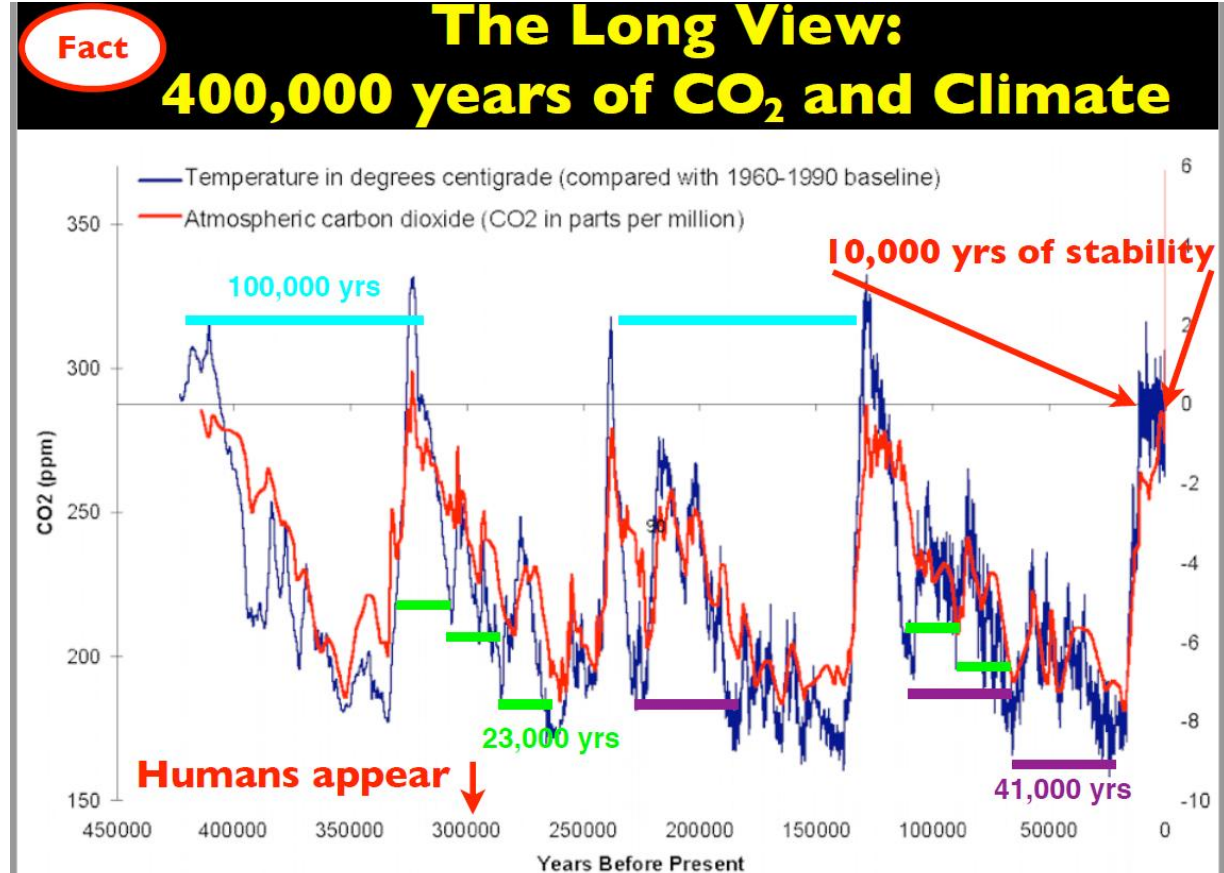
Frontiers of Science

- Orbital (Milankovic) cycles seen in past climate
 - Cannot explain rapid rise in T, CO₂ in last 50 yrs
- CO₂ and Temperature: highly correlated

Eccentricity (100,000 yr)

Obliquity (41,000 yr)

Precession (23,000 yr)

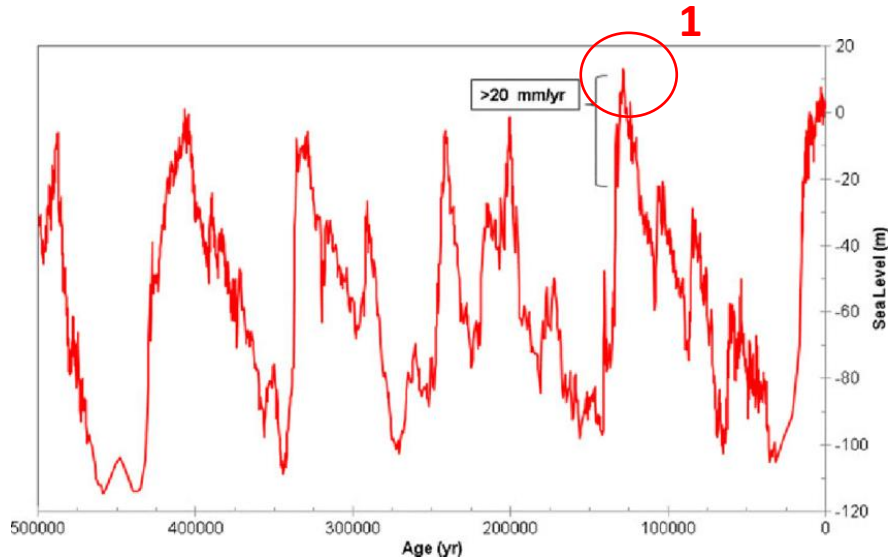




LO1: Comparing paleoclimate to today

Frontiers of Science

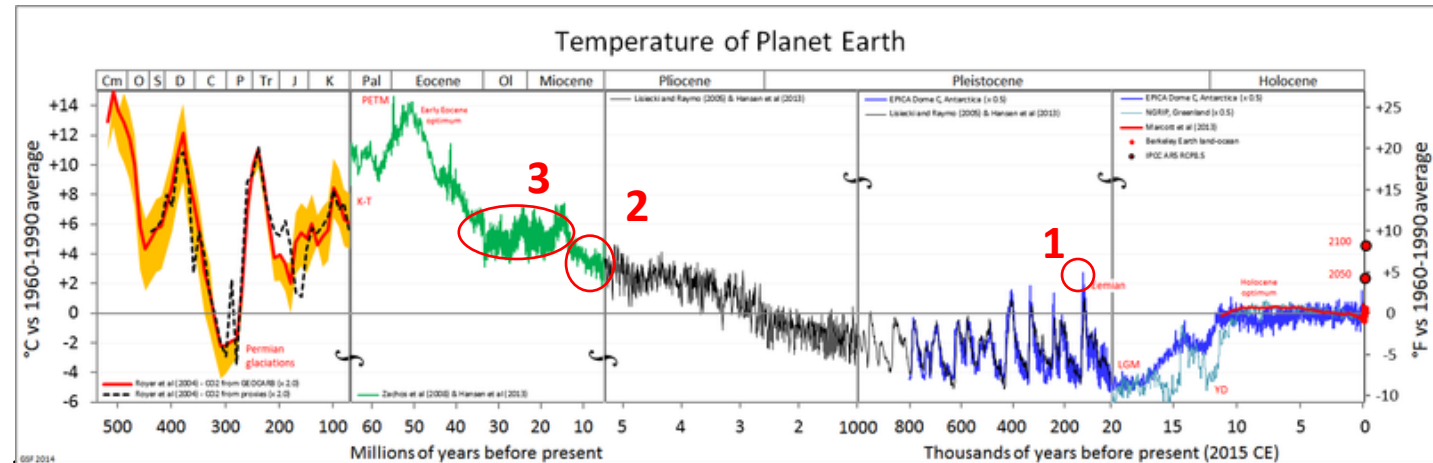
- Earth has had higher CO₂, warmer T, higher sea level (no glaciers)
- But this **rate of increase** has never been seen (except *maybe* during mass extinctions)



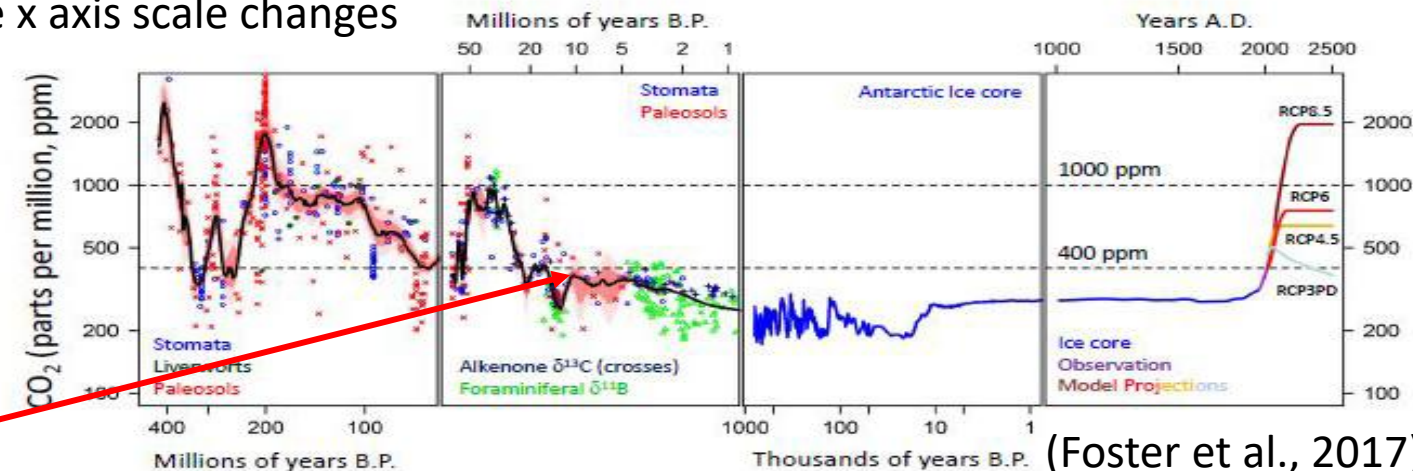
Lessons from the past:

- 1) 115-130 kyr: +2°C, 4-6 m sea level (IPCC)
- 2) 10-15 Myr: +2°C, + 6 m sea level
- 3) 20-40 Myr: +4°C, +70 m Sea level

CO₂ hasn't been above 400 ppm in 5-20 Myrs



*Note x axis scale changes

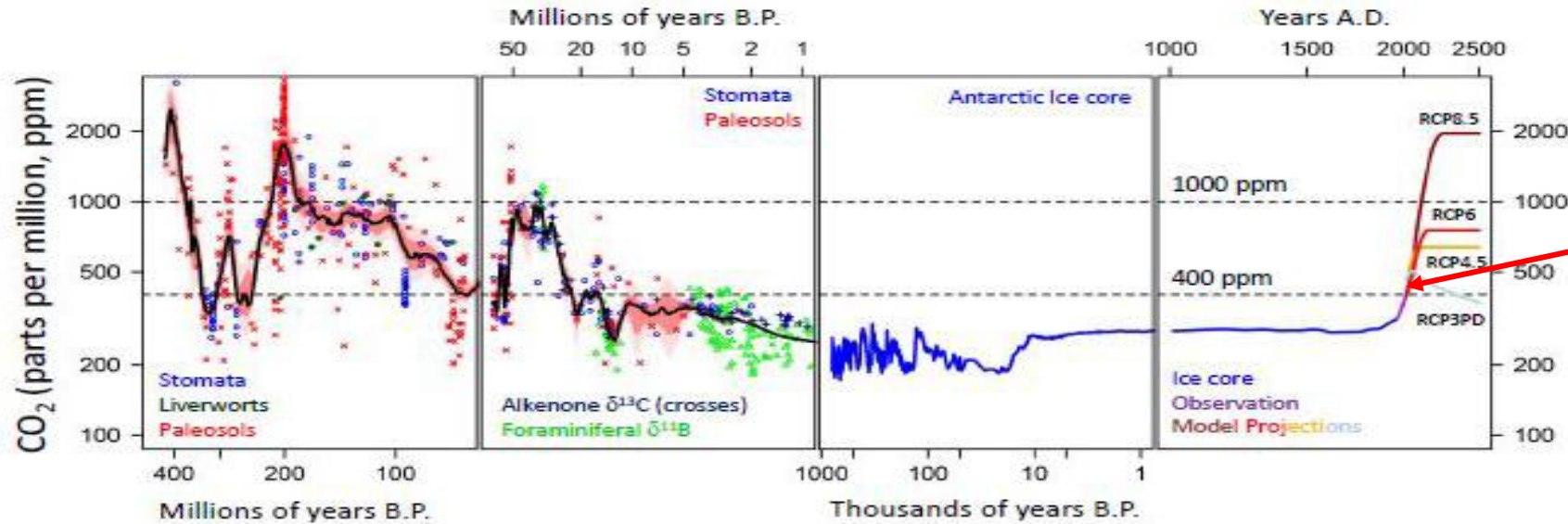


(Foster et al., 2017)



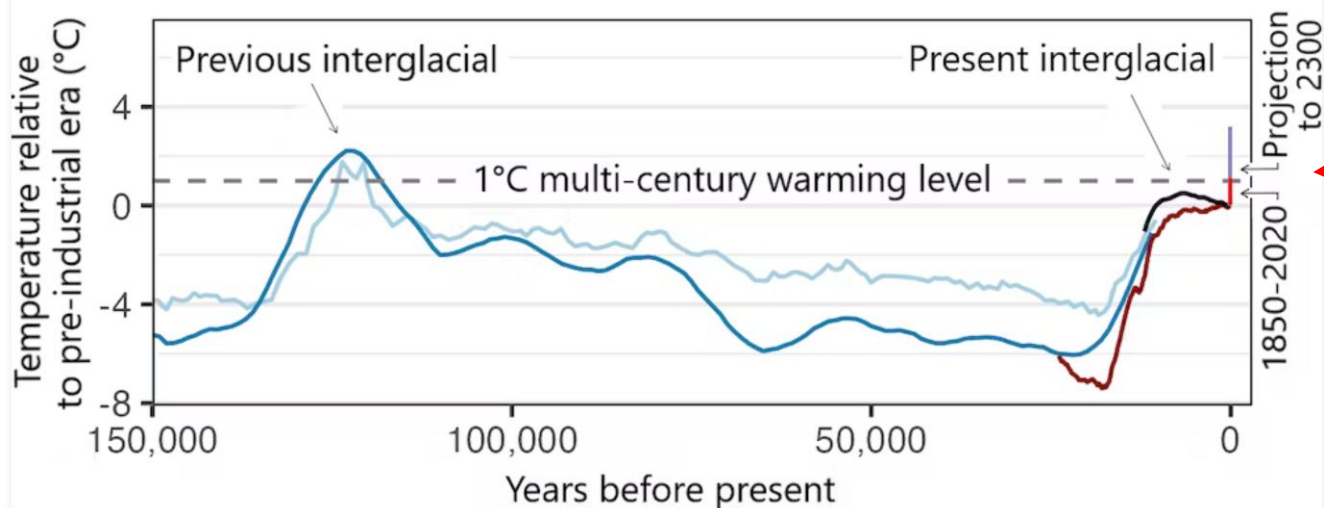
LO1: Understanding Past Climate

Frontiers of Science



The recent increase in CO₂ is higher than any time in the last few millions of years.

(Foster et al., 2017)



The average temperature increase of 1°C since 1900 is unprecedented in the past 100,000 years.

(Kaufman and McKay, 2022)



Learning Objectives

Frontiers of Science

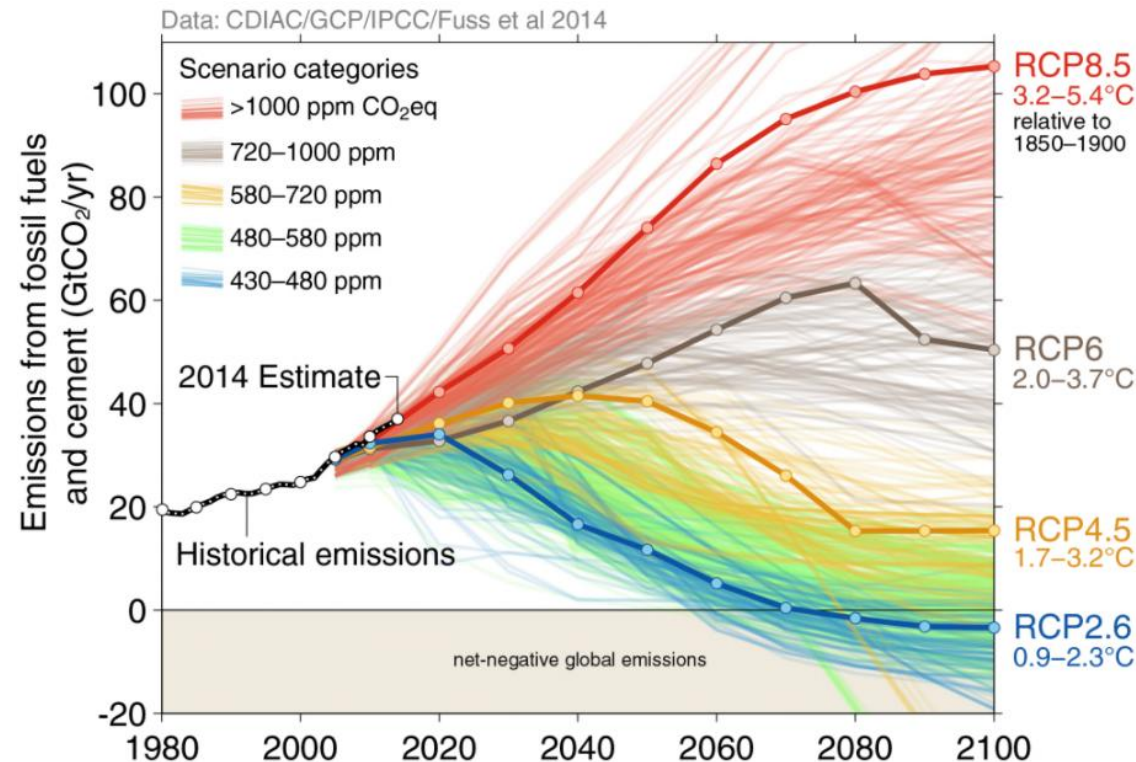
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LO2: Predicting Future Climate: Modelling

Frontiers of Science

- Climate models generalize processes in the earth system (math. models)
 - to predict how climate will change with further increases in CO₂
 - Validate model by predicting present day/past conditions (hindcasting)
- Used to understand future impacts of climate change, establish climate policy



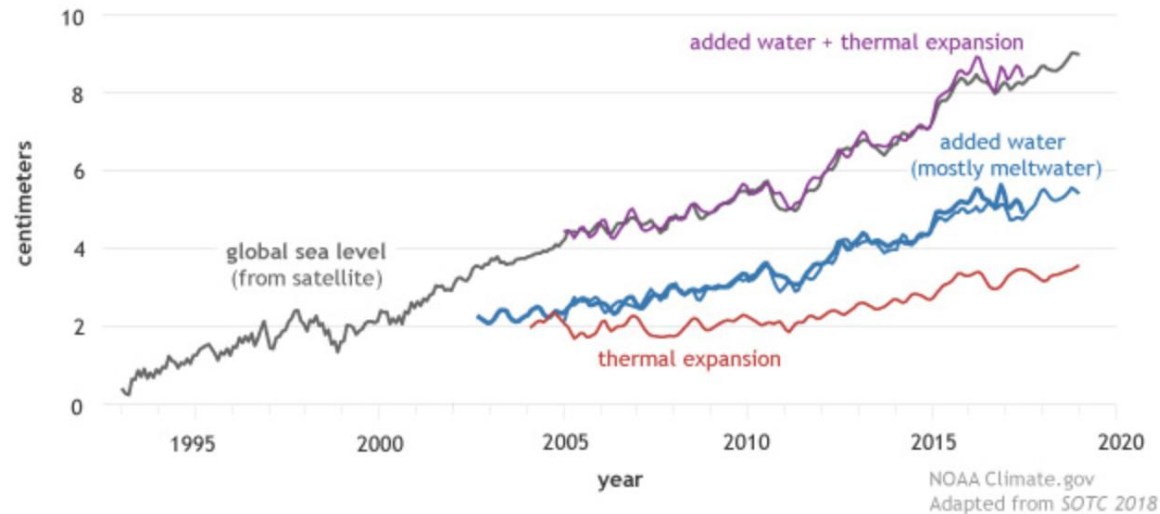


LO2: Predicting Future Climate: Modelling

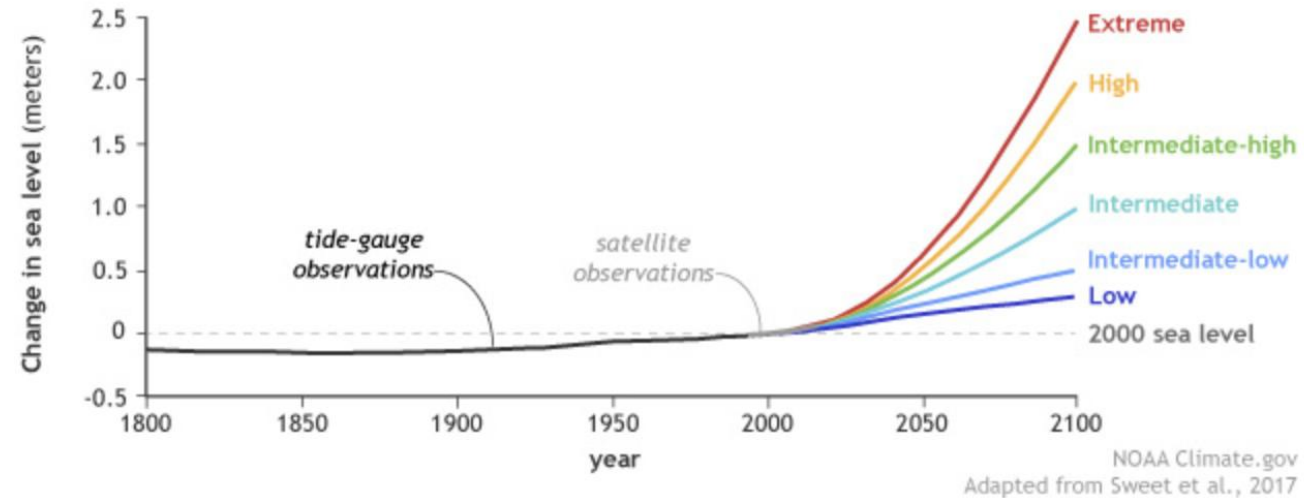
Frontiers of Science

- Modeling Sea Level Rise:
 - Melting glaciers
 - ↑ocean temperature (thermal expansion)

Contributors to global sea level rise (1993-2018)



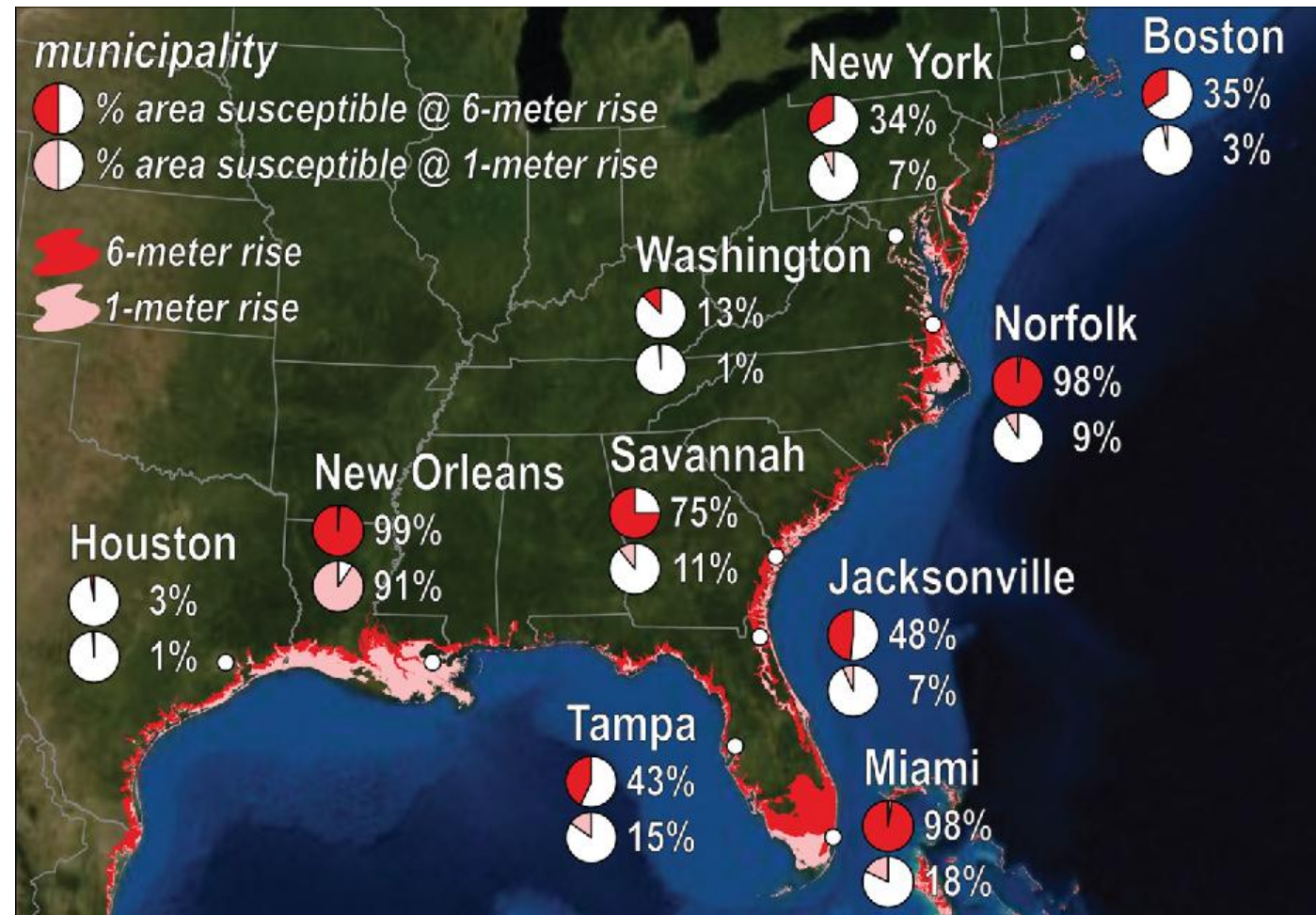
Possible future sea levels for different greenhouse gas pathways





LO-2: Modelled Effects of Sea-level Rise

Frontiers of Science

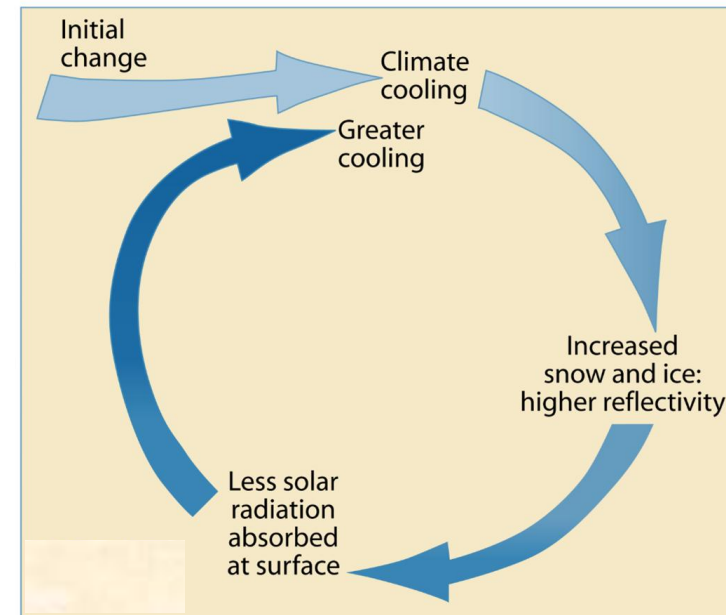
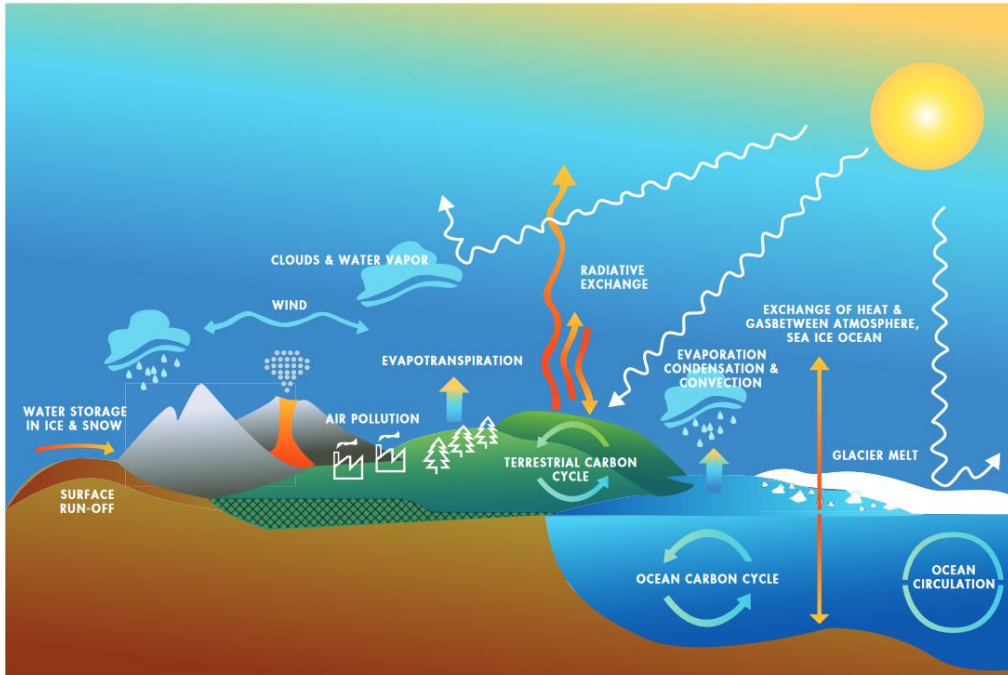


Overpeck and Weiss, 2009



L02: Feedbacks

- Feedbacks complicate the models
- Positive: effect amplifies the cause
 - $\uparrow \text{Temperature} = \downarrow \text{Arctic Ice cover} = \downarrow \text{reflected radiation} = \uparrow \text{absorbed} = \uparrow \text{ocean Temp}$
- Negative: effect suppresses the cause
 - $\uparrow \text{Temperature} = \uparrow \text{evaporation} = \uparrow \text{clouds} = \uparrow \text{reflected radiation} = \downarrow \text{Temperature}$





Learning Objectives

Frontiers of Science

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L03: Climate Fictions

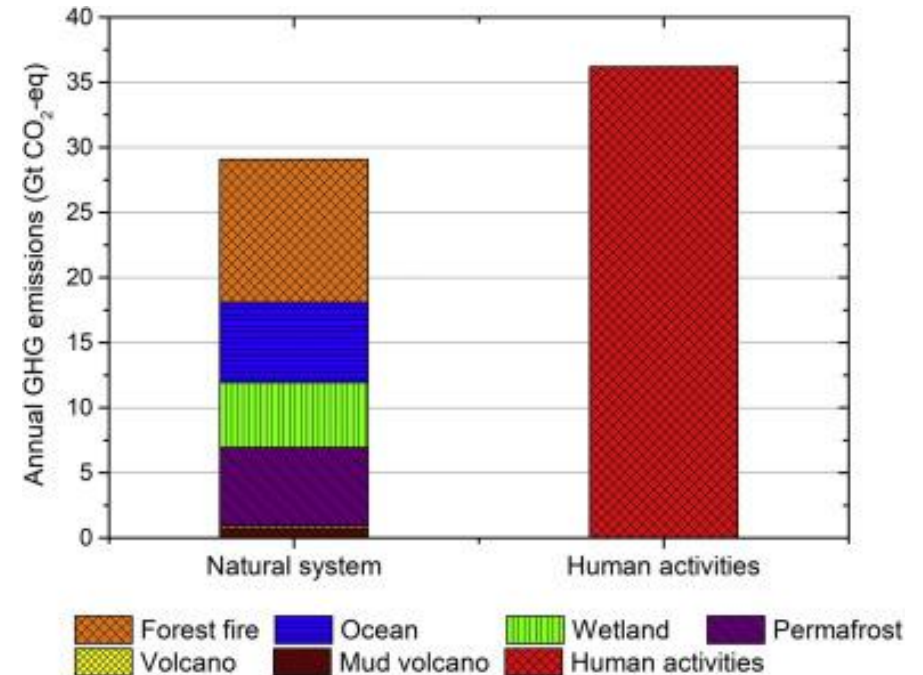
Frontiers of Science

FICTION

Natural cycles are the cause of recent global warming and increased atmospheric CO₂ concentrations

FACT

The estimated contribution of anthropogenic CO₂ emissions through fossil fuel burning is 37.3 Gigatons CO₂ per year (as of 2019)





L03: Climate Fictions

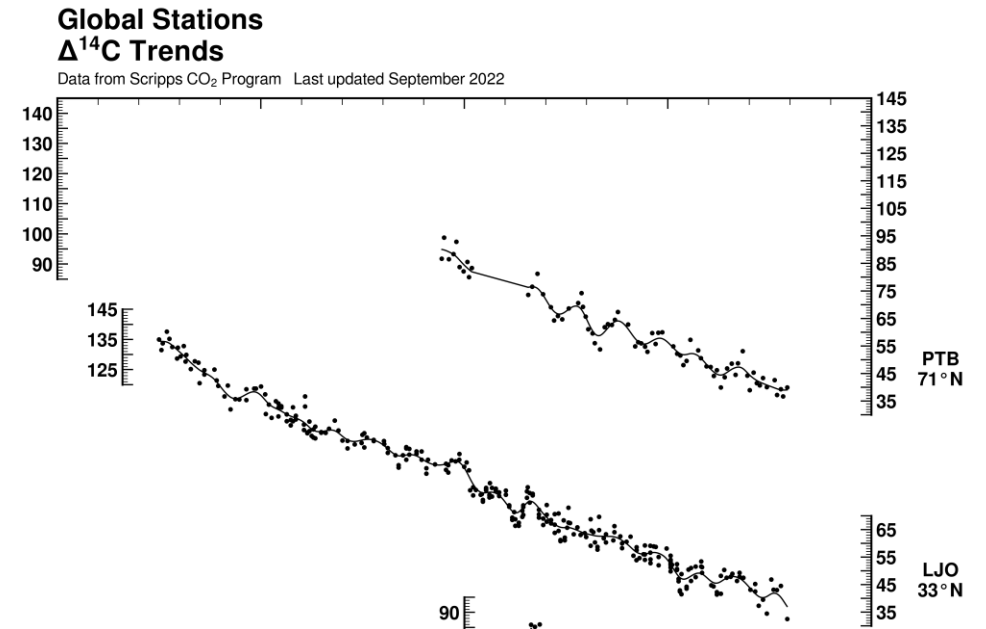
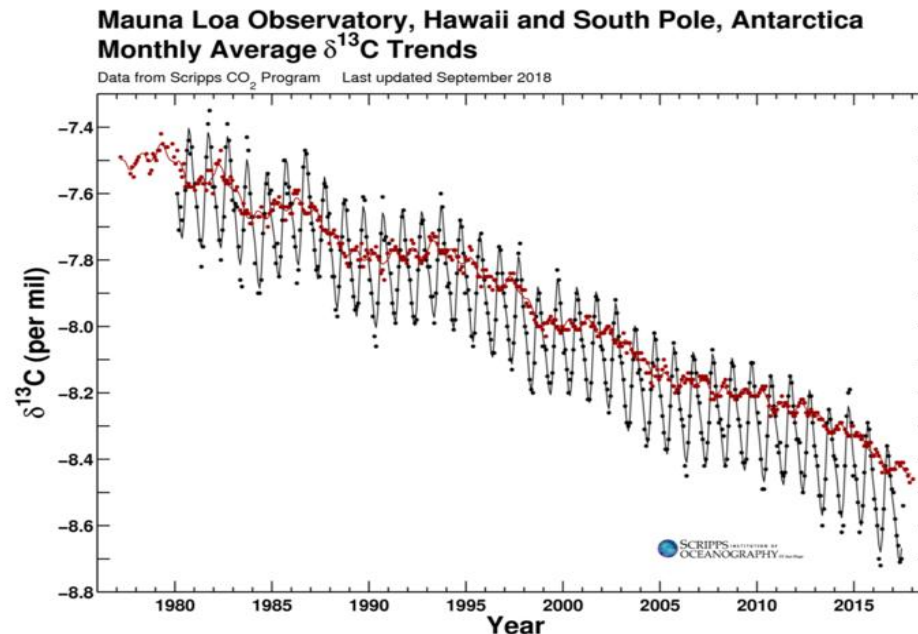
Frontiers of Science

FICTION

Natural cycles are the cause of recent global warming and increased atmospheric CO₂ concentrations

FACT

The ¹³C/¹²C ratio and ¹⁴C/¹²C of atmospheric CO₂ are both declining since the beginning of the industrial period; only fossil fuels can be the source





L03: Climate Fictions

Frontiers of Science

FICTION

Volcanoes are the cause for increased atmospheric CO₂

FACT

1. Amount of CO₂ is increasing rapidly
 - Volcanoes have **not** been erupting more frequently
 - CO₂ emissions from humans is > 60 x emissions of all volcanoes globally
2. O₂ is also decreasing, perfectly anti-correlated with increasing CO₂
 - Volcanoes emit CO₂ produced by melting carbonate sediments (doesn't use O₂)
 - Burning fossil fuels requires O₂ to burn ancient plant material
3. ¹³C/¹²C and ¹⁴C/¹²C are both decreasing
 - Volcanic CO₂ has high ¹³C: more emission wouldn't ↓ ¹³C/¹²C
 - Only fossil fuels have both low ¹³C/¹²C and low ¹⁴C/¹²C

Great 2-min Summary:

<https://www.youtube.com/watch?v=UXgDrr6qiUk&t=9s>



L03: Climate Fictions

Frontiers of Science

FICTION

Not all climate scientists agree that global warming is caused by human activity

FACT

Nearly all (99.994%) of climate scientists agree that human activities are the cause



Learning Objectives

Frontiers of Science

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LO4: Solutions

Frontiers of Science

Mitigation

- Interventions that reduce the sources of greenhouse gases (ex. carbon caps, renewable energy)
- Interventions that enhance the sinks of greenhouse gases (ex. carbon capture)
- Reduce the amount of GHGs in the atmosphere

Adaptation

- Adjustments in response to actual or expected changes in climate
- Aim to reduce harm related to global warming
- Ex. building sea walls to protect from flooding



LO4: Solutions

Frontiers of Science

Technological

Political

Economic

•

•

•



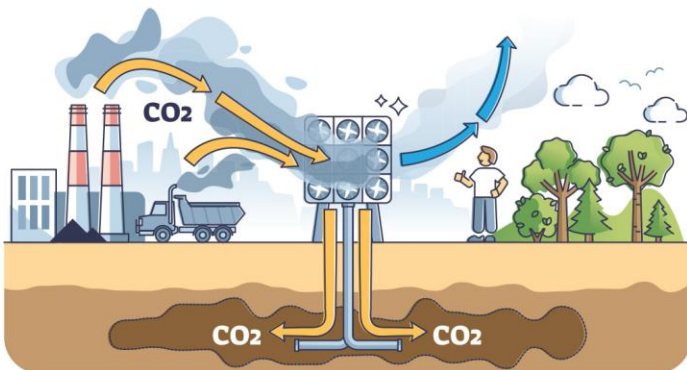
LO4: Mitigation Solutions

Frontiers of Science

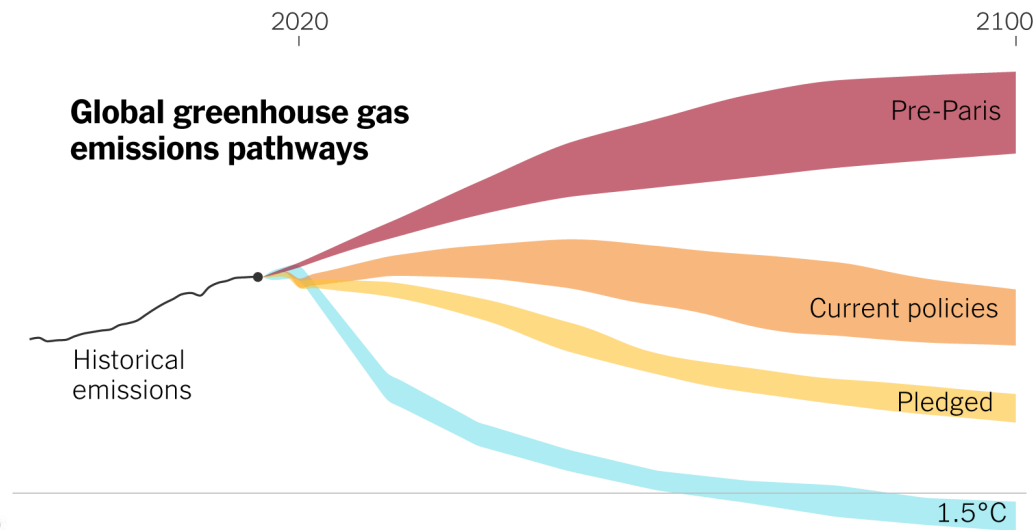
Technological



CARBON CAPTURE



Political



Economic



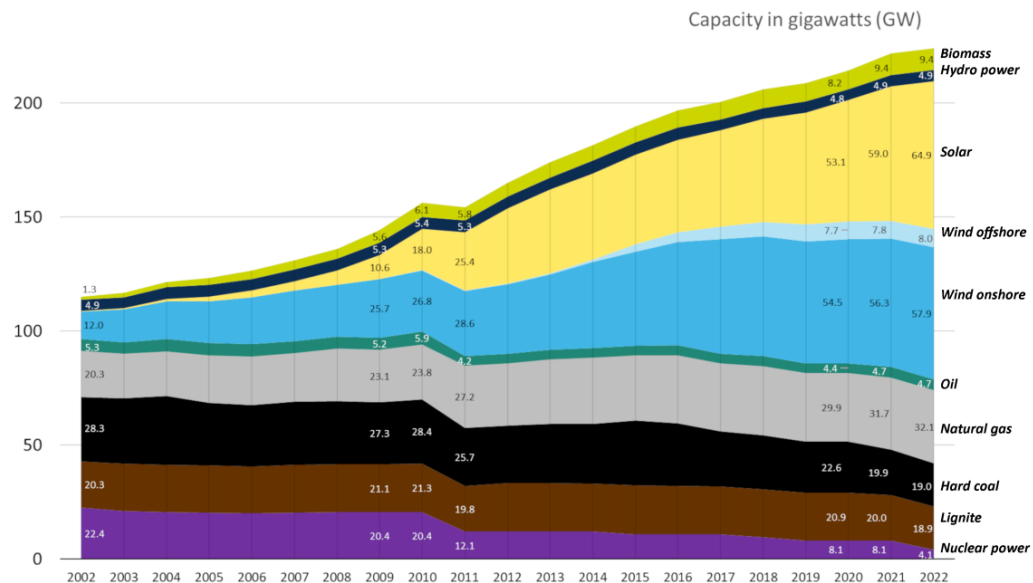


LO4: Mitigation Solutions

Frontiers of Science

Installed net power generation capacity in Germany 2002 - 2022.

Data: Fraunhofer ISE 2022 (2022 status of 19 December).



CC BY SA 4.0

Bhadla Solar Park (India) is the largest solar park in the world, with a capacity for 2.25 GW



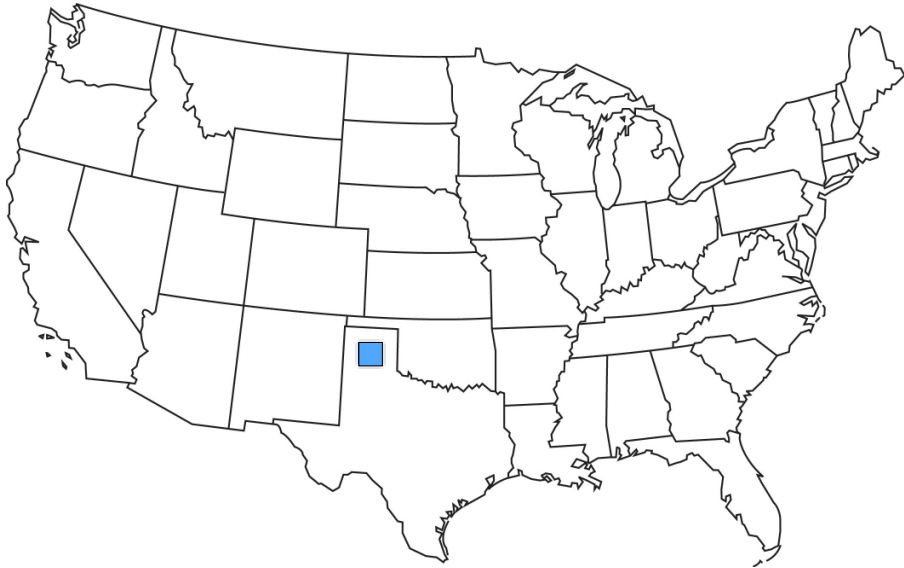
Renewable energy accounted for 52.3% of Germany's power consumption in 2023



Renewable energy is feasible!

Frontiers of Science

Solar area needed to supply US



US NREL



Bhadla Solar Park, India: largest in world
(14,000 acres ~ size of Manhattan)
Desert region is “almost unlivable” (avg. T 118F)



LO4: Mitigation Solutions

Frontiers of Science



U.S. has invested \$1.2 billion for two pioneering, large-scale carbon capture facilities that will sequester an estimated 1 million tons of CO₂ each per year

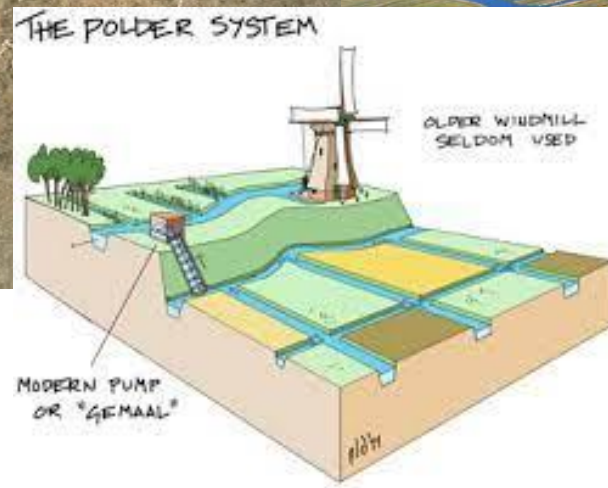


LO4: Adaptation Solutions

Frontiers of Science



A Polder and dike system is part of the Netherlands's successful approach to flood risk management





LO4: Adaptation Solutions

Frontiers of Science



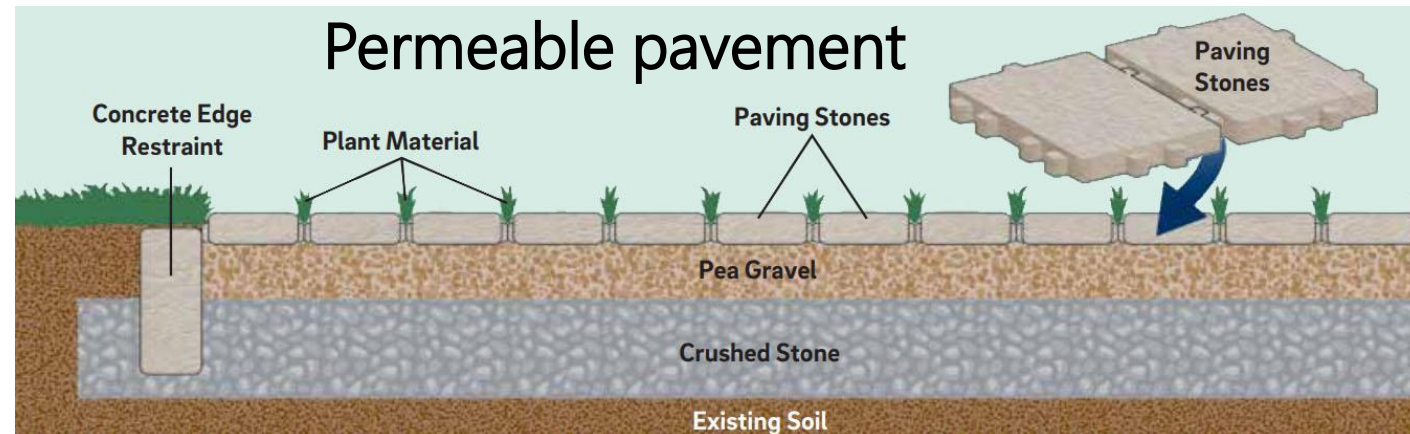
Bioswale

Green infrastructure bolsters a city's capacity to filter and absorb stormwater

Green roofs



Permeable pavement





LO4: Promise and problems of climate solutions

Frontiers of Science

- Mitigation solutions: technological, political, economic
 - Technological:
 - **Alternative energy**: solar, wind, hydrothermal, geothermal, and nuclear power
 - **Electrification/ increased efficiency** of transportation, buildings and industry
 - **Carbon capture** (requires a lot of energy currently)
 - Changing Earth's energy balance via geoengineering (sulfate aerosols= more clouds)- consequences??
 - Political:
 - Creating and adhering to or enforcing global agreements (e.g. 2015 Paris Agreement)
 - “reducing temperature rise to $<2^{\circ}\text{C}$ and adapt to new conditions that face our planet”
 - **Land use: Reduce deforestation; reduce agricultural emissions** of methane and NO_2 (fertilizer)
 - Economic:
 - Green Bonds: for development of underutilized/abandoned buildings into qualified green buildings
 - **Carbon Taxes**
 - Green technology becomes more affordable (E-Vs: 26% drop in price every year since 2010) as it becomes more popular and production increases

*Green = parameters in modelling activity